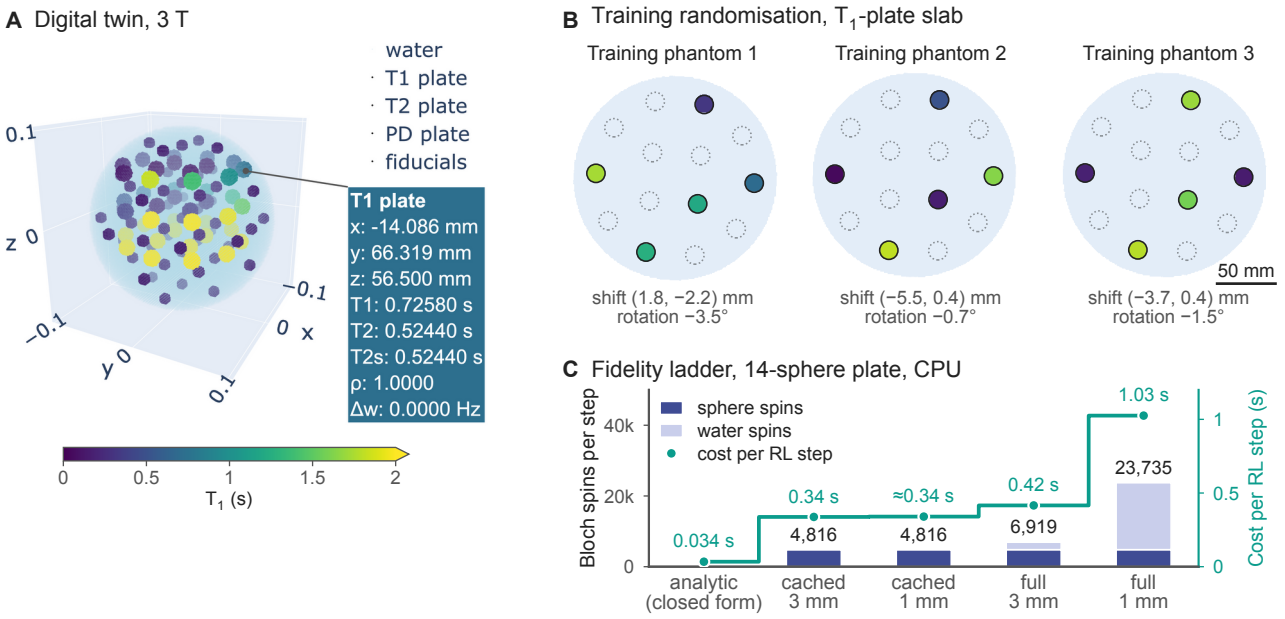

A Digital twin, 3 T
water
T1 plate
T2 plate
PD plate
fiducials
T1 plate
x: -14.086 mm
y: 66.319 mm
z: 56.500 mm
T1: 0.72580 s
T2: 0.52440 s
T2s: 0.52440 s
ρ: 1.0000
Δw: 0.0000 Hz
0.1
z 0
−0.1
y 0
0.1
−0.1
0 x
0
0.5
1
1.5
2
T1 (s)
B Training randomisation, T1-plate slab
Training phantom 1
Training phantom 2
Training phantom 3
shift (1.8, −2.2) mm
rotation −3.5°
shift (−5.5, 0.4) mm
rotation −0.7°
shift (−3.7, 0.4) mm
rotation −1.5°
50 mm
C Fidelity ladder, 14-sphere plate, CPU
sphere spins
water spins
cost per RL step
Bloch spins per step
40k
20k
0
Cost per RL step (s)
1
0.5
0
0.034 s
0.34 s
≈0.34 s
0.42 s
1.03 s
4,816
4,816
6,919
23,735
analytic
(closed form)
cached
3 mm
cached
1 mm
full
3 mm
full
1 mm